Chapter 8 — Spring Data JPA and Hibernate

Chapter 7 wrote SQL by hand. This chapter lets an **ORM** (object-relational mapper) write it. You describe Java classes and their relationships, and Hibernate generates the `SELECT`, `INSERT`, `UPDATE` and `DELETE` statements, tracks changes to your objects, and loads related objects on demand. Spring Data JPA adds **repositories**: interfaces whose methods become queries without any implementation code.

This is very productive, and it is also where many performance problems in Spring applications come from. So every important behaviour below is *measured*. The tests count the SQL statements Hibernate really sends and print them. You will see the N+1 problem happen, then disappear with three different fixes. You will also see a `LazyInitializationException`, dirty checking writing an `UPDATE` without any `save()` call, and two users colliding on an optimistic lock.

What you'll learn

- How JPA, Hibernate and Spring Data JPA relate (and which versions Boot 4.1 ships)
- Entities, ids and id generation strategies
- The persistence context: entity states, the first-level cache, dirty checking, flushing
- Relationships: `@ManyToOne`, `@OneToMany`, `@ManyToMany`, owning side, `mappedBy`, cascades, `orphanRemoval`
- Fetch types, `LazyInitializationException`, and Open Session in View
- The N+1 problem and its fixes: `JOIN FETCH`, `@EntityGraph`, batch fetching, projections
- Repositories: derived queries, `@Query` (JPQL and native), `@Modifying`, paging and sorting, projections
- Auditing, optimistic locking with `@Version`, pessimistic locking
- Schema migrations with Flyway

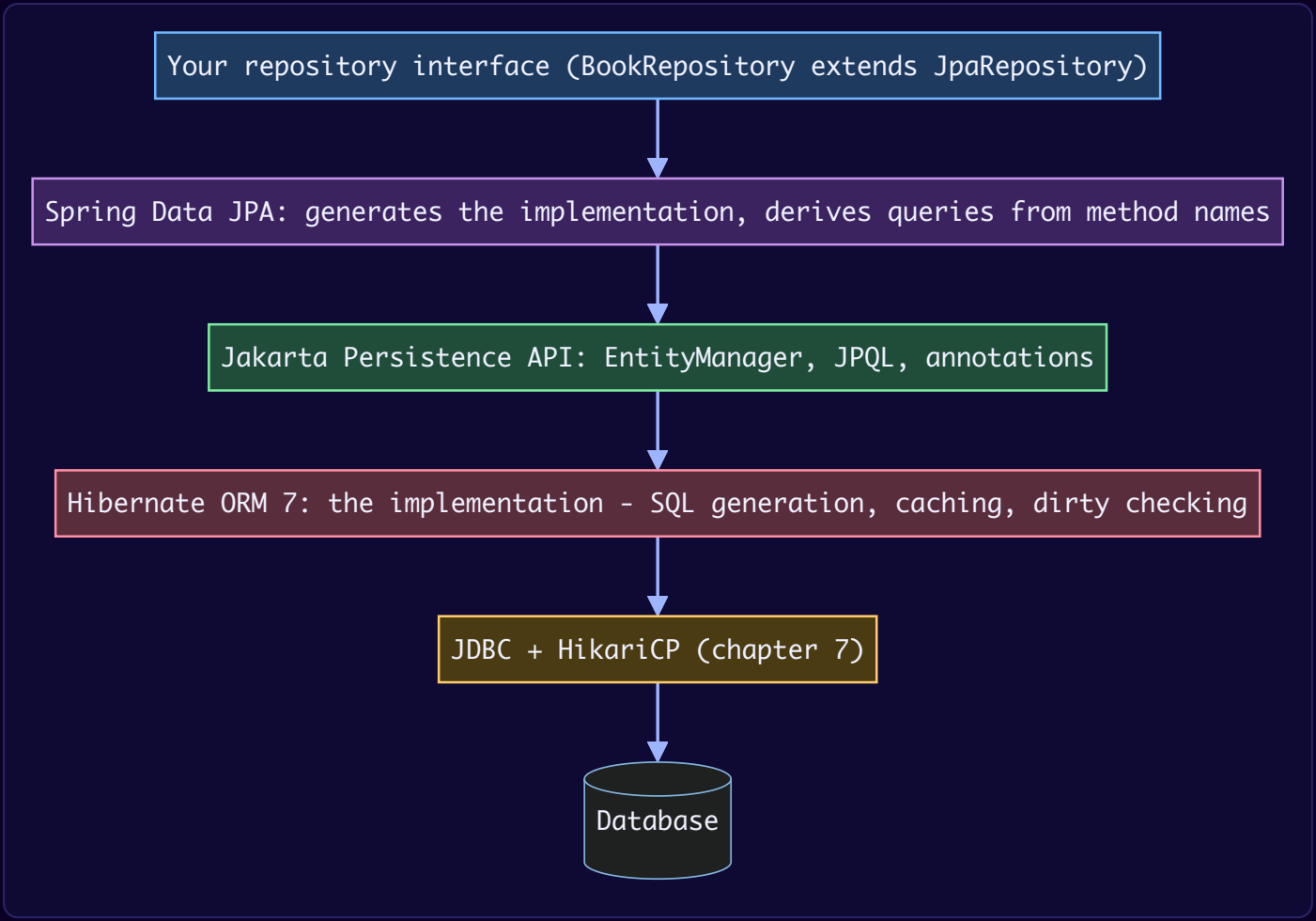
The code (Spring Boot 4.1.1 → Hibernate ORM 7.4, Spring Data 2026.0, H2 in memory):

Project	What it proves
`code/jpa-demos`	persistence context, relationships, lazy loading, N+1 with statement counts, repositories, projections, auditing, locking, open-in-view
`code/flyway-migrations`	versioned migrations + `ddl-auto=validate`

Run everything with `bash tools_run_spring.sh spring '08-*'`. The sample data is a small library: 4 authors, 8 books, 3 tags, 6 reviews (`src/main/resources/data.sql`).

8.1 JPA, Hibernate and Spring Data – who does what

What it is



Layer	What it is	Version with Boot 4.1
Jakarta Persistence (JPA)	a *specification*: the `jakarta.persistence` annotations, `EntityManager`, the JPQL query language	3.2
Hibernate ORM	the most used *implementation* of JPA. It does the real work	7.4 (Boot 4.1 upgraded to 7.4; Hibernate 7.0 came out in 2025)
Spring Data JPA	a Spring layer on top of JPA: repositories, derived queries, paging, auditing	part of the Spring Data 2026.0 release train
Spring ORM	the glue in Spring Framework: `JpaTransactionManager`, the shared `EntityManager` proxy, exception translation	Framework 7.0

A few Hibernate 7 facts worth knowing: it implements **Jakarta Persistence 3.2** and **Jakarta Data 1.0**, it needs Java 17 or later, and it is the first Hibernate version licensed under **Apache License 2.0** (earlier versions were LGPL). Spring Data 2025.1 (the Boot 4.0 generation) moved to JSpecify null annotations and added **AOT repositories**, which generate repository code at build time for faster startup and native images. Spring Data 2026.0 (Boot 4.1) added **type-safe property paths**, for example `Sort.by(Book::getPrice)` instead of `Sort.by("price")`.

When not to use JPA

JPA is a good fit for a rich domain model: many related entities, loaded, changed and saved as a unit of work. It fits poorly for:

- reporting and analytics (big joins, aggregations, window functions): use SQL through `JdbcClient` (7.3) or jOOQ

- bulk data processing (millions of rows): use JDBC batches or database-side SQL
- teams who want to see every statement: Spring Data JDBC is simpler and has no lazy loading and no persistence context (ch09)

The general ORM vs raw SQL debate is covered in SQL 14.13.

8.2 Setting up

```
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-data-jpa</artifactId>    <!-- Hibernate + Spring Data JPA + HikariCP -->
</dependency>
<dependency>
  <groupId>com.h2database</groupId>
  <artifactId>h2</artifactId>
  <scope>runtime</scope>
</dependency>
<dependency>                                                <!-- @DataJpaTest -->
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-data-jpa-test</artifactId>
  <scope>test</scope>
</dependency>
```

```
# jpa-demos/src/main/resources/application.properties
spring.jpa.hibernate.ddl-auto=create-drop
spring.jpa.defer-datasource-initialization=true          # run data.sql after Hibernate created tables
spring.jpa.properties.hibernate.generate_statistics=true # count statements
spring.jpa.properties.hibernate.session_factory.statement_inspector=ch08.jpa.support.SqlRecorder
```

Anything under `spring.jpa.properties.*` is passed to Hibernate unchanged. That is how you set any Hibernate option Boot has no property for.

`ddl-auto`: who creates the tables?

Value	Hibernate at startup	Use for
`none`	nothing	production with a migration tool
`validate`	checks that tables and columns match the entities, and fails startup if not	production with Flyway/Liquibase (see 8.16)
`update`	adds missing tables and columns, never drops	quick prototypes only. Unreliable for real schema changes
`create`	drops and creates the schema	tests
`create-drop`	like `create`, and drops again at shutdown	tests, demos (this project)

Boot's default is `create-drop` for an embedded database when no migration tool is present, and `none` otherwise.

Seeing the SQL

Option	Shows
`spring.jpa.show-sql=true`	every statement on stdout, prefixed `Hibernate:` (`@DataJpaTest` turns this on by default)
`logging.level.org.hibernate.SQL=DEBUG`	the same through the logger (better)
`logging.level.org.hibernate.orm.jdbc.bind=TRACE`	the bound parameter values too
`hibernate.generate_statistics=true`	counters: statements, entity loads, collection fetches, flushes
a `StatementInspector`	your own code sees every SQL string, which is used here to print and count

The tests use a small helper built on Hibernate's `Statistics` and the `SqlRecorder` inspector:

```
final class SqlCount {
    private final Statistics stats;
    SqlCount(EntityManagerFactory emf) { stats = emf.unwrap(SessionFactory.class).getStatistics(); }
    void reset()      { stats.clear(); SqlRecorder.clear(); }
    long statements() { return stats.getPrepareStatementCount(); }   // JDBC statements prepared
    void print(String label) { /* label + count + each recorded SQL string */ }
}
```

In your own projects you can get the same effect in tests with the *datasource-proxy* library or Hibernate statistics, and fail a test when a request uses more statements than expected. That turns N+1 bugs into test failures.

8.3 Entities and ids

What it is

An **entity** is a class whose instances are rows of a table. The JPA rules: annotate it with `@Entity`, give it an `@Id`, give it a no-argument constructor (it may be `protected`), and do not make the class or its persistent methods `final`, because Hibernate subclasses entities to build lazy proxies. **Records cannot be entities**, since they are final and immutable. Records work well as `@Embeddable` value types (Hibernate 6.2+) and as projections (8.12).

```java
@Entity
@EntityListeners(AuditingEntityListener.class)
public class Book {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String title;                // column "title" - names are mapped automatically
    private BigDecimal price;

    @ManyToOne(fetch = FetchType.LAZY, optional = false)
    @JoinColumn(name = "author_id")
    private Author author;

    @ManyToMany
    @JoinTable(name = "book_tag",
               joinColumns = @JoinColumn(name = "book_id"),
               inverseJoinColumns = @JoinColumn(name = "tag_id"))
    private Set<Tag> tags = new HashSet<>();

    @Version private Long version;       // optimistic locking, 8.14
    @CreatedDate private Instant createdAt;       // auditing, 8.13
    @LastModifiedDate private Instant updatedAt;
    @CreatedBy private String createdBy;

    protected Book() {}                  // for JPA
    public Book(String title, BigDecimal price) { this.title = title; this.price = price; }
    // getters, a few setters ...
}
```

The chapter's tests switch `show-sql` off, but running one of them with `-Dlogging.level.org.hibernate.SQL=DEBUG` shows the table Hibernate generated at startup. Boot's default naming strategy turns `createdAt` into `created_at`:

```
create table book (price numeric(38,2), author_id bigint not null, created_at timestamp(6) with time zone,
  id bigint generated by default as identity, updated_at timestamp(6) with time zone, version bigint not null,
  created_by varchar(255), title varchar(255), primary key (id))
```

Useful column annotations: `@Column(nullable = false, length = 100, unique = true)`, `@Column(precision = 10, scale = 2)` (note the default `numeric(38,2)` above), `@Enumerated(EnumType.STRING)` (always use STRING, since ORDINAL breaks when someone reorders the enum), `@Transient` (not stored), and `@Embedded` for value objects such as `Address`.

Id generation strategies

Strategy	How the id is produced	Trade-off
`IDENTITY`	database auto-increment column	simple, but Hibernate must `INSERT` **immediately** on `persist` to learn the id, which also rules out JDBC insert batching
`SEQUENCE`	database sequence. Hibernate fetches blocks of ids (`allocationSize`, default 50)	best for PostgreSQL/Oracle: inserts can wait until flush and be batched
`AUTO` (default)	Hibernate picks, which in Hibernate 6+ is usually a sequence per entity, `book_seq`	know what you get
`UUID` (JPA 3.1+)	`java.util.UUID` generated in Java	no round trip, safe to generate anywhere. 16 bytes, and random UUIDs spread index inserts
assigned (no `@GeneratedValue`)	you set it	natural keys. `save()` must then use `merge`, see 8.8

This project uses IDENTITY so `data.sql` can insert rows without ids. You can see the immediate inserts in 8.5.

equals and hashCode

Entities live in `Set`s and are compared across persistence contexts, so these methods matter:

- The **default** (`Object` identity) is correct inside one persistence context, because Hibernate guarantees one instance per row there (8.4). This project relies on that.
- **Never** base `hashCode` on a generated id that is `null` before persist. The hash changes after saving, and the object gets lost inside a `HashSet`.
- Good options: a natural business key (ISBN), or `equals` on id with a **constant** `hashCode` (`getClass().hashCode()`).
- Never include lazy associations in `equals`, `hashCode` or `toString`. Doing so triggers loading, or a `LazyInitializationException`.

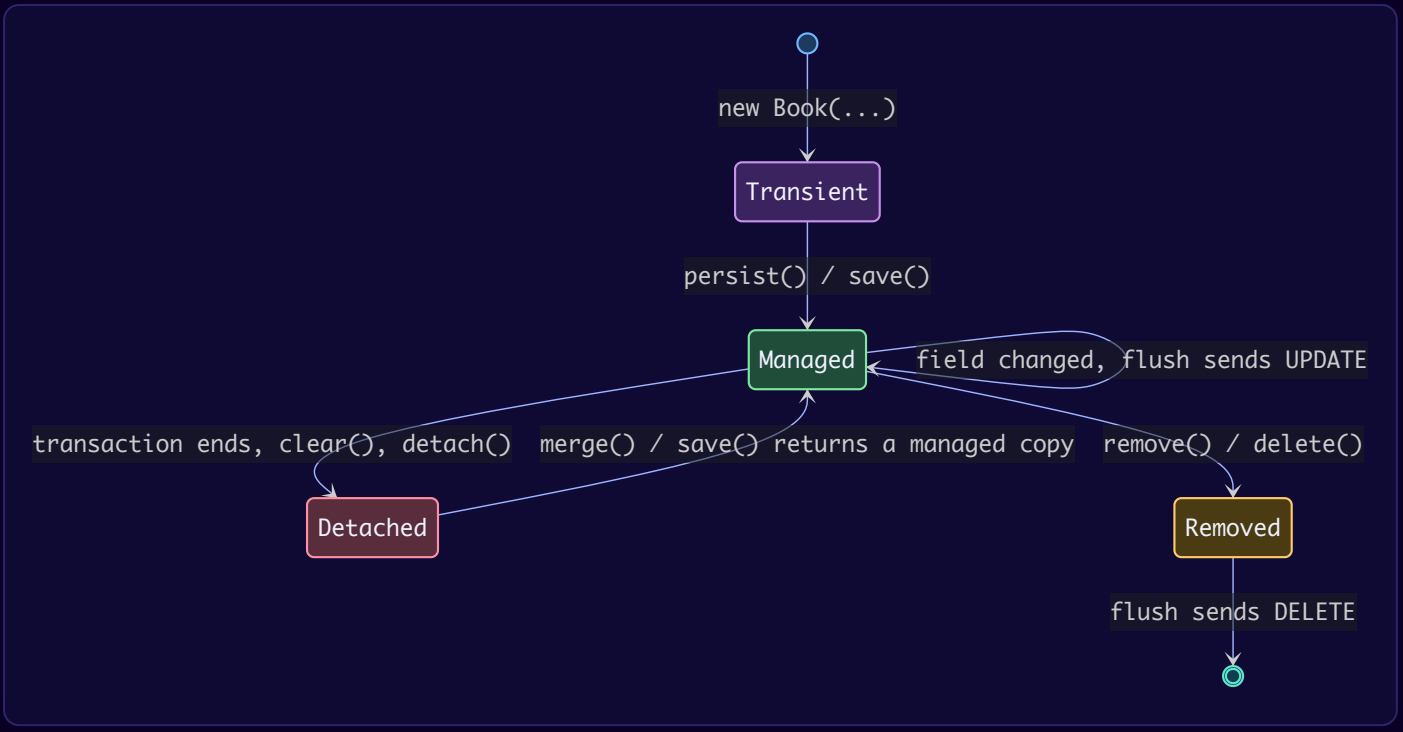
8.4 The persistence context and dirty checking

What it is

The **persistence context** is Hibernate's in-memory workspace for one unit of work. It is the `EntityManager` (Hibernate calls it `Session`). In Spring it usually lives exactly as long as the transaction. It does three jobs:

1. **Identity map / first-level cache**: each row is loaded at most once and always gives the *same* Java object.
2. **Change tracking (dirty checking)**: at load time it keeps a snapshot of each entity. At **flush** time it compares every managed entity with its snapshot and writes an `UPDATE` for the ones that changed. You never call `update()`.
3. **Write-behind**: inserts, updates and deletes are queued and sent at flush. Flush happens before commit, before a query that might read the changed tables, or when you call `flush()`.

Entity states



State	Tracked?	Example
Transient	no — Hibernate does not know it	`new Book(...)`
Managed	yes — changes are flushed automatically	anything returned by `find`, a query, or `persist` inside the current transaction
Detached	no — it has an id but no persistence context	an entity returned by a repository call *after* its transaction ended
Removed	scheduled for `DELETE` at flush	after `remove()`

Demo — one row, one object, one SELECT

```
Book first = books.findById(1L).orElseThrow();
Book second = books.findById(1L).orElseThrow();
System.out.println("same Java object? " + (first == second));
```

```
same Java object? true
findById(1) twice -> 1 SQL statement(s)
    select
b1_0.id,b1_0.author_id,b1_0.created_at,b1_0.created_by,b1_0.price,b1_0.title,b1_0.updated_at,b1_0.version from
book b1_0 where b1_0.id=?
```

The second `findById` came from the persistence context without any SQL. (`@DataJpaTest` wraps each test in one transaction, and so in one persistence context.)

Demo — dirty checking: an UPDATE without save()

```
Book book = books.findById(1L).orElseThrow();
book.setPrice(new BigDecimal("14.99"));   // just change the Java object...
em.flush();                               // ...Hibernate compares with its snapshot and writes SQL
```

```
setPrice() + flush, no save() -> 2 SQL statement(s)
    select
b1_0.id,b1_0.author_id,b1_0.created_at,b1_0.created_by,b1_0.price,b1_0.title,b1_0.updated_at,b1_0.version from
book b1_0 where b1_0.id=?
    update book set author_id=?,created_at=?,created_by=?,price=?,title=?,updated_at=?,version=? where id=?
and version=?
version is now 1
```

In real code the flush happens at commit. `LibraryService.changePrice` is exactly this, and it is tested in 8.13:

```java
@Transactional
public void changePrice(Long bookId, BigDecimal newPrice) {
    books.findById(bookId).orElseThrow().setPrice(newPrice);   // no save()
}
```

Notice that the `UPDATE` lists **all** columns. By default Hibernate uses one prepared `UPDATE` per entity type, which it can cache. `@DynamicUpdate` on the entity makes it send only the changed columns instead.

Demo – detached entities are not tracked

```java
Book book = books.findById(1L).orElseThrow();
em.clear();                              // persistence context emptied: book is DETACHED
book.setPrice(new BigDecimal("1.00"));
em.flush();
```

```
change a detached entity + flush -> 1 SQL statement(s)
    select ... from book b1_0 where b1_0.id=?
```

No `UPDATE` was sent: the change is simply lost. To store changes to a detached entity you call `save()` (which calls `merge`). Merge loads the current row, copies your state onto the managed copy, and returns that copy. Use the return value.

readOnly transactions and JPA

`@Transactional(readOnly = true)` tells Spring to put the Hibernate session in read-only mode with manual flushing. Hibernate then keeps no snapshots and never runs dirty checking, which saves memory and CPU when you load many entities just to display them. Spring Data's `SimpleJpaRepository` is annotated `@Transactional(readOnly = true)` at class level and plain `@Transactional` on its write methods.

Common mistakes

- Calling `save()` on an entity you just loaded in the same transaction. It is harmless but unnecessary, because the entity is managed already.
- Ignoring the result of `save()` / `merge()` for a detached entity, then modifying the old detached object.
- Loading 100 000 entities in one transaction. The persistence context keeps all of them *and* their snapshots. Process in pages and call `clear()`, or use a read-only transaction, projections or JDBC.

In the interview

30-second answer: "The persistence context is the `EntityManager`'s first-level cache and unit of work. It returns the same object for the same row, keeps a snapshot of every managed entity, and at flush it writes `UPDATE`s for entities that changed. That is dirty checking, and it is why you don't need `save()` for loaded entities. Entities are transient, managed, detached or removed. Only managed ones are tracked."

- *When does Hibernate flush?* → Before commit, before a query that touches affected tables (AUTO flush mode), and on an explicit `flush()`.

- *persist vs merge?* → `persist` makes a transient object managed (the same instance). `merge` copies a detached object's state onto a managed copy and returns the copy.

Self-check

1. How many SELECTs does `findById(1)` twice cost in one transaction? → One.
2. Why was no UPDATE sent after `em.clear()`? → The entity was detached, so nothing tracked it.
3. What does `readOnly = true` save with Hibernate? → Snapshots and dirty checking, and flushes.

8.5 Relationships

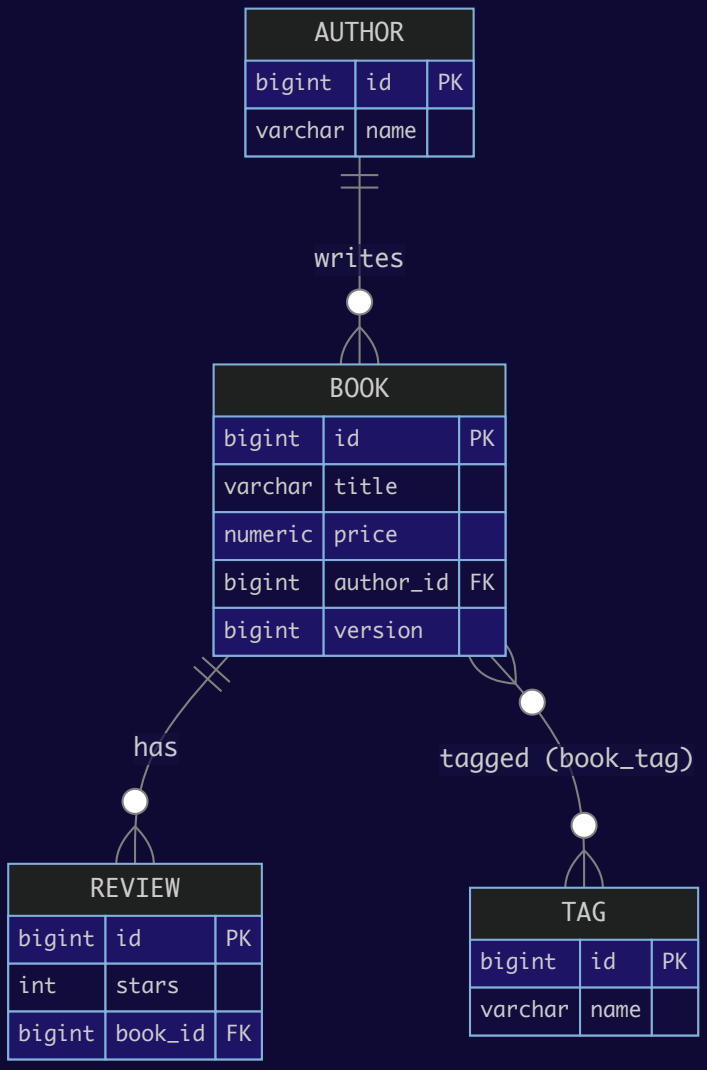
What it is

Annotation	Meaning	Default fetch	Example
`@ManyToOne`	many of these point to one of those	**EAGER**	`Book.author`
`@OneToMany`	one of these has a collection of those	LAZY	`Author.books`
`@OneToOne`	exactly one on each side	**EAGER**	`User.profile`
`@ManyToMany`	join table between the two	LAZY	`Book.tags`

The owning side and mappedBy

In the database a relationship is **one foreign-key column** (or a join table). In Java it may be navigable from both sides, but only one side may control that column. That side is the **owning side**:

- For `@ManyToOne` / `@OneToMany` the owner is the **many** side (`Book.author` → `book.author_id`).
- The other side says `mappedBy = "author"`, meaning "I am just a mirror of `Book.author`". This is the **inverse side**.
- JPA **only reads the owning side** when it writes SQL. Adding a book to `author.getBooks()` without setting `book.author` stores nothing.



That is why `Author` has helper methods that keep both sides in sync:

```
@Entity
public class Author {
    @Id @GeneratedValue(strategy = GenerationType.IDENTITY) private Long id;
    private String name;

    @OneToMany(mappedBy = "author", cascade = CascadeType.ALL, orphanRemoval = true)
    private List<Book> books = new ArrayList<>();

    public void addBook(Book book)    { books.add(book);    book.setAuthor(this); }
    public void removeBook(Book book) { books.remove(book); book.setAuthor(null); }
    // ...
}
```

Cascades

A **cascade** passes an operation from parent to children:

CascadeType	Passes on
`PERSIST`	saving a new parent also saves new children
`MERGE`	merging the parent merges the children
`REMOVE`	deleting the parent deletes the children
`REFRESH`, `DETACH`	reload / detach together
`ALL`	all of the above

Use cascades only from an **aggregate root** to the parts it owns (author → books in this demo, order → order lines in real life). Never cascade `REMOVE` on `@ManyToMany` or towards shared entities. Deleting a book must not delete the "sci-fi" tag that other books use.

```
Author author = new Author("N. K. Jemisin");
author.addBook(new Book("The Fifth Season", new BigDecimal("11.00")));
author.addBook(new Book("The Obelisk Gate", new BigDecimal("11.00")));
authors.save(author);                    // persist cascades to both books
```

```
save(new author with 2 new books) -> 3 SQL statement(s)
    insert into author (name,id) values (?,default)
    insert into book (author_id,created_at,created_by,price,title,updated_at,version,id) values
(?,?,?,?,?,?,?,default)
    insert into book (author_id,created_at,created_by,price,title,updated_at,version,id) values
(?,?,?,?,?,?,?,default)
ids assigned: author 5, books [9, 10]
```

The inserts ran **during** `save()`, before any flush. That is the IDENTITY strategy at work (8.3).

orphanRemoval

`orphanRemoval = true` means *"a child removed from the collection should be deleted"*. `CascadeType.REMOVE` only deletes children when the *parent* is deleted.

```
Author pratchett = authors.findById(2L).orElseThrow();
Book smallGods = pratchett.getBooks().stream()
        .filter(b -> b.getTitle().equals("Small Gods")).findFirst().orElseThrow();
pratchett.removeBook(smallGods);          // no repository.delete() call
em.flush();
```

```
remove 'Small Gods' from author.books + flush -> 4 SQL statement(s)
    select a1_0.id,a1_0.name from author a1_0 where a1_0.id=?
    select
b1_0.author_id,b1_0.id,b1_0.created_at,b1_0.created_by,b1_0.price,b1_0.title,b1_0.updated_at,b1_0.version from
book b1_0 where b1_0.author_id=?
    delete from book_tag where book_id=?
    delete from book where id=? and version=?
```

Hibernate first removed the book's rows from the `book_tag` join table (Book owns that many-to-many), then deleted the book. The first version of this demo removed a book that had **reviews**, and the database rejected the delete with a foreign-key violation (`REVIEW FOREIGN KEY(BOOK_ID) REFERENCES BOOK(ID)`). Hibernate only cleans up relationships it knows it owns. `Review.book` points at the book from the other direction, so you must deal with those rows yourself.

Many-to-many advice

A plain `@ManyToMany` suits a pure link with no data of its own. As soon as the link needs data ("tagged by whom, when?"), turn the join table into an **entity** (`BookTag` with two `@ManyToOne`s). Use `Set`, not `List`, for

many-to-many. Hibernate handles `List` without an order column (a "bag") inefficiently: removing one element deletes all the link rows and re-inserts the rest.

Common mistakes

- Setting only the inverse side (`author.getBooks().add(book)`), so nothing is saved.
- `cascade = ALL` everywhere, which deletes shared data.
- Bidirectional mappings everywhere. Map the reverse side only when you really navigate it.
- `@OneToMany` without `mappedBy`. Hibernate then creates an extra join table.

In the interview

> **30-second answer:** "The owning side is the one holding the foreign key, in a one-to-many that is the `@ManyToOne` side, and it is the only side JPA reads when writing. The other side uses `mappedBy`. Keep both sides in sync with helper methods. Cascade from aggregate roots only. `orphanRemoval` deletes children removed from the collection, while `CascadeType.REMOVE` deletes children when the parent is deleted."

Self-check

1. Which side owns `Author.books` ↔ `Book.author`? → `Book.author` (the `author_id` column).
2. Why were two DELETEs needed to remove one book? → The `book_tag` link rows go first, then the book row.
3. `orphanRemoval` vs `CascadeType.REMOVE`? → Removal from the collection vs deletion of the parent.

8.6 Fetch types and LazyInitializationException

What it is

- **LAZY**: the related data is loaded only when you first touch it. For a collection, Hibernate puts a special `PersistentBag` / `PersistentSet` in the field. For a single entity it puts a generated **proxy** subclass there.
- **EAGER**: the related data is loaded together with the owner, always.

Make everything LAZY and decide per use case what to load (see 8.7). The JPA defaults are EAGER for `@ManyToOne` and `@OneToOne`, so change them explicitly: `@ManyToOne(fetch = FetchType.LAZY)`. EAGER cannot be switched off for a single query, while LAZY can always be fetched eagerly when you need it.

LazyInitializationException

A lazy collection can only load itself while its persistence context is open. Outside a transaction, a repository call opens and closes its own short transaction. The entity it returns is already **detached**:

```
@SpringBootTest(webEnvironment = SpringBootTest.WebEnvironment.NONE)   // no test transaction
class LazyLoadingTest {
    @Test
    void lazyCollectionOutsideTheTransaction() {
        Author a = authors.findById(2L).orElseThrow();   // transaction opens and closes inside findById
        assertThatThrownBy(() -> a.getBooks().size())
                .isInstanceOf(LazyInitializationException.class);
    }
}
```

```
LazyInitializationException: Cannot lazily initialize collection of role 'ch08.jpa.Author.books' with key '2'
(no session)
```

"No session" means no open persistence context. The fixes:

```java
@EntityGraph(attributePaths = "books")
Optional<Author> findWithBooksById(Long id);       // load the books in the same query

@Transactional(readOnly = true)                     // or: do the work inside a transaction
public AuthorSummary summary(Long authorId) {
    Author a = authors.findById(authorId).orElseThrow();
    return new AuthorSummary(a.getName(), a.getBooks().size());   // return a DTO, not the entity
}
```

```
fetched with @EntityGraph -> 3 books
inside @Transactional service -> AuthorSummary[name=Terry Pratchett, books=3]
```

Fix	Good?
fetch what the use case needs (JOIN FETCH / `@EntityGraph`)	✅ best
do the work in a `@Transactional` service and return a DTO	✅ best
a DTO projection query (8.12)	✅ for read-only views
`spring.jpa.open-in-view=true`	⚠️ hides the problem, see 8.15
switch the mapping to EAGER	❌ every query pays for it
`hibernate.enable_lazy_load_no_trans=true`	❌ each lazy load opens its own mini-transaction: N+1 with no consistency

---

8.7 The N+1 problem

What it is

You run **1** query to load N parents. Then, for each parent, touching a lazy association fires **1 more** query. With N = 1000 that is 1001 round trips for data that one query could return. It is the most common JPA performance bug, and a functional test cannot see it, because the result is correct, only slow.

Demo – watch it happen

```java
List<Author> all = authors.findAll();
all.forEach(a -> System.out.println("  " + a.getName() + ": " + a.getBooks().size() + " books"));
```

```
  Ursula K. Le Guin: 2 books
  Terry Pratchett: 3 books
  Octavia E. Butler: 1 books
  Isaac Asimov: 2 books
findAll() then a.getBooks().size() -> 5 SQL statement(s)
    select a1_0.id,a1_0.name from author a1_0
    select
b1_0.author_id,b1_0.id,b1_0.created_at,b1_0.created_by,b1_0.price,b1_0.title,b1_0.updated_at,b1_0.version from
book b1_0 where b1_0.author_id=?
    select b1_0.author_id,b1_0.id,... from book b1_0 where b1_0.author_id=?
    select b1_0.author_id,b1_0.id,... from book b1_0 where b1_0.author_id=?
    select b1_0.author_id,b1_0.id,... from book b1_0 where b1_0.author_id=?
```

1 + 4 = 5 statements. The test asserts `statements() == 1 + all.size()`.

EAGER does not fix it — demo

`Review.book` was left at the JPA default (`@ManyToOne` = EAGER). We never touch `review.getBook()`:

```
var all = reviews.findAll();
```

```
reviews.findAll() with EAGER @ManyToOne Review.book (6 reviews) -> 6 SQL statement(s)
    select r1_0.id,r1_0.book_id,r1_0.stars from review r1_0
    select b1_0.id,b1_0.author_id,... from book b1_0 where b1_0.id=?
    select b1_0.id,b1_0.author_id,... from book b1_0 where b1_0.id=?
    select b1_0.id,b1_0.author_id,... from book b1_0 where b1_0.id=?
    select b1_0.id,b1_0.author_id,... from book b1_0 where b1_0.id=?
    select b1_0.id,b1_0.author_id,... from book b1_0 where b1_0.id=?
```

1 query for the reviews, then 1 per distinct book (5), **even though the code never used the books**. EAGER tells Hibernate that the association must be loaded. It does not say *how*. For JPQL and derived queries Hibernate runs the query as written, then loads each eager association separately.

Fix 1 — JOIN FETCH

```
@Query("select a from Author a join fetch a.books order by a.id")
List<Author> findAllWithBooks();
```

```
JOIN FETCH -> 1 SQL statement(s)
    select a1_0.id,b1_0.author_id,b1_0.id,...,a1_0.name from author a1_0 join book b1_0 on
a1_0.id=b1_0.author_id order by a1_0.id
```

Hibernate 6+ removes the duplicate parent rows that a join produces on its own, so `distinct` is no longer needed. Note that it is an **inner** join: authors with no books would disappear from the result. Use `left join fetch` to keep them.

Fix 2 — @EntityGraph

```
@EntityGraph(attributePaths = "books")
@Query("select a from Author a order by a.id")
List<Author> findAllWithBooksGraph();
```

```
@EntityGraph(attributePaths = "books") -> 1 SQL statement(s)
    select a1_0.id,b1_0.author_id,b1_0.id,...,a1_0.name from author a1_0 left join book b1_0 on
a1_0.id=b1_0.author_id order by a1_0.id
```

The effect is the same, but it is a **left** join and needs no JPQL. It also works on derived query methods (`findWithBooksById` above). Entity graphs can also be declared on the entity with `@NamedEntityGraph`.

Fix 3 — batch fetching

Same code as the N+1 demo, one property added:

```
@DataJpaTest(showSql = false, properties = "spring.jpa.properties.hibernate.default_batch_fetch_size=16")
```

```
findAll() + books with default_batch_fetch_size=16 -> 2 SQL statement(s)
    select a1_0.id,a1_0.name from author a1_0
    select b1_0.author_id,b1_0.id,... from book b1_0 where b1_0.author_id in (?,?,?,?,?,?,?,?,?,?,?,?,?,?,?,?)
```

When the first lazy collection is touched, Hibernate loads the collections of up to 16 authors in the persistence context at once. N+1 becomes 1 + ⌈N/16⌉. It needs no query changes, which makes it a good **global**

safety net. Per association, use `@BatchSize(size = 16)`.

Paging a JOIN FETCH (Hibernate 7.4)

Paging and fetching a collection in the same query was a classic trap. `LIMIT 2` on a joined result limits *rows*, not authors, so older Hibernate versions loaded everything and paged in memory, with the warning `HHH90003004: firstResult/maxResults specified with collection fetch; applying in memory`. Hibernate 7.4 pages the parents in a subquery instead. Our test, run on Boot 4.1.1:

```
@Query(value = "select a from Author a join fetch a.books",
       countQuery = "select count(a) from Author a")
Page<Author> pageWithBooks(Pageable pageable);

authors.pageWithBooks(PageRequest.of(0, 2, Sort.by("id")));
```

```
JOIN FETCH + Pageable(page 0, size 2) -> 2 SQL statement(s)
    select a1_0.id,b1_0.author_id,...,a1_0.name from (select a1_0.id,a1_0.name from author a1_0 where
exists(select 1 from book b1_0 where a1_0.id=b1_0.author_id) order by a1_0.id offset ? rows fetch first ? rows
only) a1_0(id,name) join book b1_0 on a1_0.id=b1_0.author_id order by a1_0.id
    select count(a1_0.id) from author a1_0
```

The inner subquery picks 2 authors with `offset/fetch first`, and the outer query joins their books. On older Hibernate versions, set `hibernate.query.fail_on_pagination_over_collection_fetch=true` so the in-memory case fails loudly. Then page the ids first and fetch the collections in a second query (or rely on batch fetching).

Choosing a fix

Fix	Queries	Good for	Watch out
`JOIN FETCH`	1	one known use case	inner join by default. Fetching two `List` collections at once → `MultipleBagFetchException` (use `Set`, or two queries)
`@EntityGraph`	1	the same, declared on a repository method	same cartesian-product risk with several collections
batch size	1 + N/size	global default, several collections	still more than one query
DTO projection	1	read-only screens and APIs	no entities, so no dirty checking (usually what you want)

Fetching two collections in one query multiplies rows: 10 books × 10 tags = 100 rows per author. For several collections, prefer batch fetching or separate queries.

Common mistakes

- Returning entities from a REST controller. Jackson walks every getter, triggering lazy loads (N+1) or `LazyInitializationException`, and exposes your schema. Return DTOs.
- Using `toString()`/Lombok `@Data` on entities with associations. Printing loads them.
- Testing with 3 rows. N+1 only hurts with real volumes, so assert statement counts in tests instead.

In the interview

30-second answer: "N+1 is one query for the parents plus one per parent when a lazy association is touched, or even untouched with EAGER to-one mappings. I find it by counting statements with Hibernate statistics or a datasource proxy. I fix it per use case with `JOIN FETCH` or an `@EntityGraph`, globally reduce it with `default_batch_fetch_size`, or avoid entities with a DTO projection. EAGER is not a fix, it makes the problem permanent."

- *Why doesn't EAGER prevent N+1?* → EAGER says *whether* to load, not *how*. For queries Hibernate loads the associations with separate selects after the main query.

- *JOIN FETCH with pagination?* → Historically in-memory pagination (HHH90003004). Hibernate 7.4 pages parents in a subquery. Otherwise page ids first, then fetch.
- *`MultipleBagFetchException`?* → Two `List` collections fetched in one query. Use `Set`, or fetch them in separate queries.

Self-check

1. How many statements does `findAll()` + touching books cost for 4 authors without a fix? With batch size 16? → 5, and 2.
2. Which fix keeps authors that have no books? → `@EntityGraph` (left join), or `left join fetch`.
3. Why did `reviews.findAll()` run 6 statements? → EAGER `@ManyToOne` loaded each distinct book separately.

TL;DR

Mappings LAZY. Decide what to fetch per query with JOIN FETCH / `@EntityGraph`. `default_batch_fetch_size` as a safety net. DTO projections for read views. Count statements in tests.

8.8 Repositories

What it is

A **repository** is an interface. At startup Spring Data creates a proxy that implements it, using `SimpleJpaRepository` for the standard methods and generated query code for your own ones.

```
public interface BookRepository extends JpaRepository<Book, Long> { ... }
```

Interface	Adds
`Repository<T, ID>`	marker only. Declare exactly the methods you want
`CrudRepository` / `ListCrudRepository`	`save`, `findById`, `findAll`, `count`, `delete...` (the List variant returns `List` instead of `Iterable`)
`PagingAndSortingRepository` / `ListPagingAndSortingRepository`	`findAll(Sort)`, `findAll(Pageable)`
`JpaRepository`	all of the above + `flush`, `saveAndFlush`, `deleteAllInBatch`, `getReferenceById`, ...
`JpaSpecificationExecutor<T>`	dynamic `Specification` (criteria) queries

Expose only what callers should use. A `Repository` with a handful of methods is often better than handing out `deleteAll()`.

How save() decides between INSERT and UPDATE

`save(entity)` asks "is this entity new?". If yes, it calls `persist` (the same instance becomes managed). If not, it calls `merge` (it returns a **different**, managed instance). An entity counts as new when its id is `null`, or when a wrapper-type `@Version` field is `null`. With **assigned ids** every entity has an id, so `save` runs `merge`, which first `SELECT`s the row. Implement `Persistable<ID>` to tell Spring Data yourself.

Transactions in repositories

`SimpleJpaRepository` methods are transactional: read methods `readOnly = true`, write methods read-write. Each call **outside** a service transaction is its own tiny transaction, which leads straight to the "wrong layer" bug of 7.13. Put `@Transactional` on the service method so that all repository calls join one transaction.

8.9 Derived queries

The method name *is* the query. Spring Data parses it at startup, so a typo fails startup, not a request:

```
List<Book> findByTitleContainingIgnoreCase(String part);
List<Book> findByPriceLessThanOrderByPriceAsc(BigDecimal max);
List<Book> findByAuthorName(String authorName);          // traverses book.author.name (a join)
long countByAuthorName(String authorName);
boolean existsByTitle(String title);
List<Book> findTop2ByOrderByPriceDesc();
List<Book> findByTagsName(String tag, Sort sort);        // many-to-many join + dynamic sort
```

```
containing 'go'      : [Small Gods, Going Postal]
price < 10 asc       : [I, Robot, Foundation, Guards! Guards!, A Wizard of Earthsea]
by author name       : [A Wizard of Earthsea, The Dispossessed]
count by author      : 3
exists 'Kindred'     : true
top 2 by price       : [Kindred, The Dispossessed]
tag sci-fi, title asc: [Foundation, I, Robot, Kindred, The Dispossessed]
tag fantasy, by price: [Guards! Guards!, A Wizard of Earthsea, Small Gods]
```

The last line used the Spring Data 2026.0 type-safe form `Sort.by(Book::getPrice)`. A renamed getter is then a compile error rather than a runtime one. The SQL Spring Data generated for some of them:

```
containing / count / exists -> 3 SQL statement(s)
    select b1_0.id,...,b1_0.version from book b1_0 where upper(b1_0.title) like upper(?) escape '\'
    select count(b1_0.id) from book b1_0 join author a1_0 on a1_0.id=b1_0.author_id where a1_0.name=?
    select b1_0.id from book b1_0 where b1_0.title=? fetch first ? rows only
```

`existsBy` fetches just one id. Note also that `upper(title) like` cannot use a normal index on `title` (SQL 12.13).

Keywords

Keyword	Example	SQL idea
`And`, `Or`	`findByTitleAndPrice`	`title = ? and price = ?`
`Is`, `Equals`, `Not`	`findByTitleNot`	`title <> ?`
`LessThan`, `GreaterThanEqual`, `Between`	`findByPriceBetween(a, b)`	`price between ? and ?`
`IsNull`, `IsNotNull`	`findByAuthorIsNull`	`author_id is null`
`Like`, `StartingWith`, `EndingWith`, `Containing`	`findByTitleStartingWith`	`like 'x%'`
`IgnoreCase`	`findByTitleIgnoreCase`	`upper(title) = upper(?)`
`In`, `NotIn`	`findByIdIn(List)`	`id in (...)`
`True`, `False`	`findByActiveTrue`	`active = true`
`OrderBy...Asc/Desc`	`findByAuthorNameOrderByTitleDesc`	`order by`
`First`, `Top`, `TopN`, `Distinct`	`findFirst3ByOrderByPriceAsc`	`fetch first 3 rows`
nested property	`findByAuthorName`	join to `author`
prefixes	`count...By`, `exists...By`, `delete...By`	count / exists / load-then-delete

Special parameters: `Sort`, `Pageable`, `Limit` (Spring Data 3.2+), `ScrollPosition` (keyset scrolling, 3.1+). Return types: `List`, `Optional`, `Stream` (close it), `Page`, `Slice`, `Window`, a single entity, or a projection.

When not to use derived queries: when the name grows past two or three conditions (`findByAuthorNameAndPriceLessThanAndTagsNameOrderBy...`). Switch to `@Query`.

8.10 @Query: JPQL, native SQL and @Modifying

JPQL

JPQL (Jakarta Persistence Query Language) looks like SQL but talks about **entities and fields**, not tables and columns: `b.author.name` instead of a join written by hand.

```
@Query("select b from Book b where b.author.name = :name and b.price <= :max order by b.price")
List<Book> byAuthorUpTo(@Param("name") String name, @Param("max") BigDecimal max);
```

```
JPQL  Pratchett <= 11 : [Guards! Guards!, Small Gods]
```

(`@Param` can be omitted when you compile with `-parameters`, which the Boot parent POM enables.)

Native SQL

```
@Query(value = "SELECT title FROM book WHERE price > ?1 ORDER BY title", nativeQuery = true)
List<String> titlesAbove(BigDecimal min);
```

```
native titles > 11    : [Going Postal, Kindred, The Dispossessed]
```

Use native queries for database features JPQL lacks (window functions, `jsonb` operators, CTEs, `ON CONFLICT`). The price is that the query is tied to one database and to table and column names.

Bulk updates with @Modifying

```
@Modifying(clearAutomatically = true) // bulk UPDATE bypasses the persistence context -> clear it
@Query("update Book b set b.price = b.price * :factor where b.author.id = :authorId")
int reprice(@Param("authorId") Long authorId, @Param("factor") BigDecimal factor);
```

```
@Modifying bulk update -> 1 SQL statement(s)
    update book b1_0 set price=(b1_0.price*cast(? as numeric(38,2))) where b1_0.author_id=?
bulk update rows      : 2 -> Asimov now [BookRow[title=Foundation, price=15.98], BookRow[title=I, Robot,
price=13.98]]
```

A bulk statement goes straight to the database. Entities already loaded in the persistence context keep their **old** values, and `@Version` is **not** incremented. `clearAutomatically = true` empties the context afterwards, and `flushAutomatically = true` flushes pending changes before the statement runs. `@Modifying` methods need a read-write transaction.

Dynamic queries

For search screens with optional filters, write `Specification`s (`JpaSpecificationExecutor`), Query by Example, or Querydsl, instead of dozens of derived methods. All three build the query at runtime from the filters that are present.

Common mistakes

- Building JPQL with string concatenation. That is injection again: always use parameters.
- Forgetting that JPQL uses entity names (`Book`, `b.author.name`), not table names.
- Bulk updates followed by reading "stale" entities from the same persistence context.

8.11 Pagination and sorting

```
Page<Book> findByPriceGreaterThan(BigDecimal min, Pageable pageable);   // also runs a COUNT query
Slice<Book> findByAuthorIdIn(List<Long> authorIds, Pageable pageable);  // no COUNT, fetches size+1 rows

Page<Book> page = books.findByPriceGreaterThan(new BigDecimal("8"),
        PageRequest.of(1, 2, Sort.by("price").descending()));     // page index 1 = the second page
Slice<Book> slice = books.findByAuthorIdIn(List.of(1L, 2L), PageRequest.of(0, 3, Sort.by("id")));
```

```
Page 1 of 3 (6 rows total): [Going Postal, Small Gods]
Slice 0: [A Wizard of Earthsea, The Dispossessed, Guards! Guards!], hasNext=true
Page + Slice -> 3 SQL statement(s)
    select b1_0.id,...,b1_0.version from book b1_0 where b1_0.price>? order by b1_0.price desc offset ? rows
fetch first ? rows only
    select count(*) from book b1_0 where b1_0.price>?
    select b1_0.id,...,b1_0.version from book b1_0 where b1_0.author_id in (?,?) order by b1_0.id fetch first
? rows only
```

(Column lists shortened with `...` here and in a few other listings.) The `Slice` query asks for `size + 1` rows: if the extra row arrives, `hasNext()` is true.

Return type	Extra query	Knows	Use for
`Page<T>`	`COUNT(*)`	total elements and pages	UIs with page numbers
`Slice<T>`	none (fetches one extra row)	whether there is a next slice	"load more" / infinite scroll
`Window<T>` + `ScrollPosition`	none	next position	keyset (cursor) scrolling over big tables
`List<T>` + `Limit`	none	nothing	"top N"

Pages are numbered from **0**. `Page` and `Slice` use `OFFSET`, which gets slower the deeper you page, because the database still reads and throws away the skipped rows. For large tables use keyset pagination (`Window` with `ScrollPosition.keyset()`). The reasoning is in SQL 12.14 – OFFSET vs keyset pagination. Always sort by something unique, such as `id` as a tie-breaker, or rows can repeat or vanish between pages.

In Spring MVC, a `Pageable` controller parameter is filled from `?page=0&size=20&sort=price,desc` (ch05).

8.12 Projections

What it is

A **projection** returns only the fields you need, as an interface or a record, not the whole entity. There is no persistence context overhead, no dirty checking and no lazy loading surprises, and the SELECT names fewer columns.

```java
public interface BookSummary {          // interface projection (closed: only plain getters)
    String getTitle();
    BigDecimal getPrice();
}
public record BookRow(String title, BigDecimal price) {}   // class-based (DTO) projection

List<BookSummary> findByAuthorNameOrderByTitle(String authorName);      // interface
List<BookRow> findByPriceLessThan(BigDecimal max);                       // record
@Query("select new ch08.jpa.BookRow(b.title, b.price) from Book b where b.author.id = :id order by b.title")
List<BookRow> rowsForAuthor(@Param("id") Long authorId);                 // JPQL constructor expression
<T> List<T> findByAuthorId(Long authorId, Class<T> type);               // dynamic: caller picks the type
```

```
interface projection SQL -> 1 SQL statement(s)
    select b1_0.title,b1_0.price from book b1_0 join author a1_0 on a1_0.id=b1_0.author_id where a1_0.name=?
order by b1_0.title
interface projection: Going Postal 11.50
interface projection: Guards! Guards! 8.99
interface projection: Small Gods 10.99
record projection     : [BookRow[title=Foundation, price=7.99], BookRow[title=I, Robot, price=6.99]]
constructor expression: [BookRow[title=A Wizard of Earthsea, price=9.99], BookRow[title=The Dispossessed,
price=12.50]]
dynamic (BookSummary) : Kindred
```

Only `title` and `price` were selected, compared with eight columns for the entity.

Kind	How	Notes
closed interface	getters that match properties	optimized SELECT of just those columns. Nested interfaces for associations
open interface	`@Value("#{target.title + ' by ' + target.author.name}")`	loads the full entity, so no SELECT optimization
class / record (DTO)	constructor parameters match properties	derived queries select only those columns. With `@Query` use `select new ...`
dynamic	`<T> List<T> find...(..., Class<T> type)`	one method, several shapes
native query	interface projection with column aliases	`select title as title, ...`

Rule of thumb: load entities when you are going to *change* them, and use projections when you are going to *show* them.

8.13 Auditing

What it is

Auditing fills "who and when" columns automatically: `@CreatedDate`, `@LastModifiedDate`, `@CreatedBy`, `@LastModifiedBy`.

```
@SpringBootApplication
@EnableJpaAuditing                        // switch it on
public class JpaDemosApplication {
    @Bean
    AuditorAware<String> auditor() {      // "who is the current user?"
        return () -> Optional.of("system");   // real app: read it from Spring Security's context
    }
}

@Entity
@EntityListeners(AuditingEntityListener.class)   // on each audited entity
public class Book { @CreatedDate Instant createdAt; @LastModifiedDate Instant updatedAt; @CreatedBy String
createdBy; ... }
```

This test runs without a test transaction, so each step commits:

```
Book b = books.findById(bookId).orElseThrow();
library.changePrice(bookId, new BigDecimal("10.50"));   // @Transactional, dirty checking, no save()
Book after = books.findById(bookId).orElseThrow();
```

```
created: by=system at=true version=0
after changePrice: price=10.50 version=1 updatedAt moved? true
```

The auditing listener runs on the JPA `@PrePersist` / `@PreUpdate` callbacks, so it only sees changes that go through the persistence context. Bulk `@Modifying` updates bypass it. For a full history of every change use Hibernate Envers (`@Audited`, revision tables) or database-level change data capture.

8.14 Locking: optimistic and pessimistic

Optimistic locking with @Version

The problem: Alice and Bob both open the edit page for the same book, which is version 0. Alice saves. Bob saves a few seconds later and silently overwrites Alice's change. This is the *lost update* from 7.11 and SQL 13.7, now spread over two HTTP requests, where no database lock can help.

How it works:

1. Add a `@Version` field (`Long` / `Integer`, or a timestamp).
2. Every `UPDATE` becomes `... SET ..., version = version + 1 WHERE id = ? AND version = ?`, using the version the entity was loaded with.
3. If another transaction changed the row in the meantime, **0 rows** match. Hibernate reports it, and Spring throws `ObjectOptimisticLockingFailureException` (an `OptimisticLockingFailureException`).

```
Book aliceCopy = books.findById(bookId).orElseThrow();   // two users load version 0
Book bobCopy = books.findById(bookId).orElseThrow();

aliceCopy.setPrice(new BigDecimal("8.00"));
books.save(aliceCopy);                                   // merge -> UPDATE ... WHERE id=? AND version=0

bobCopy.setPrice(new BigDecimal("12.00"));
books.save(bobCopy);                                     // still version 0 -> 0 rows -> exception
```

```
both loaded version 0 and 0
Alice saves -> 2 SQL statement(s)
    select
b1_0.id,b1_0.author_id,b1_0.created_at,b1_0.created_by,b1_0.price,b1_0.title,b1_0.updated_at,b1_0.version from
book b1_0 where b1_0.id=?
    update book set author_id=?,created_at=?,created_by=?,price=?,title=?,updated_at=?,version=? where id=?
and version=?
Bob saves -> ObjectOptimisticLockingFailureException: Row was already updated or deleted by another
transaction for entity [ch08.jpa.Book with id '10']
price in DB = 8.00 (Alice's)
```

Alice's `save` is a `merge` of a detached entity, so it first `SELECT`s the current row, then updates with the version check. For Bob the merge found version 1 in the database, while his object said 0, so it failed.

Handling it: tell the user "someone else changed this, reload", which a REST API reports as `409 Conflict`. Or reload and retry automatically when the change is safe to reapply. In REST APIs, send the version to the client, for example as an `ETag`, and require it back in `If-Match` (ch05).

Pessimistic locking

Lock the row in the database for the rest of the transaction:

```
@Lock(LockModeType.PESSIMISTIC_WRITE)
Optional<Book> findWithLockById(Long id);
```

```
@Lock(PESSIMISTIC_WRITE) -> 1 SQL statement(s)
    select
b1_0.id,b1_0.author_id,b1_0.created_at,b1_0.created_by,b1_0.price,b1_0.title,b1_0.updated_at,b1_0.version from
book b1_0 where b1_0.id=? for update
```

Other transactions that want the same row wait until this one commits. Set a lock timeout (the `jakarta.persistence.lock.timeout` query hint) so they do not wait forever. How `FOR UPDATE`, `NOWAIT` and `SKIP LOCKED` behave is covered in SQL 13.12.

	Optimistic (`@Version`)	Pessimistic (`@Lock`)
Conflict detected	at write time (0 rows updated)	prevented. Others wait
Cost	none until a conflict	locks held, possible deadlocks and waiting
Works across requests / user think time	✅	❌ (a lock lives only as long as its transaction)
Best when	conflicts are rare (most web apps)	conflicts are frequent and short (counters, stock, job queues)

In the interview

30-second answer: "Optimistic locking adds a version column. Updates include `WHERE version = ?`, and if another transaction got there first, zero rows match and Spring throws `ObjectOptimisticLockingFailureException`, which I turn into a 409 or a retry. It holds no locks, so it works across user think time. Pessimistic locking uses `SELECT ... FOR UPDATE` through `@Lock(PESSIMISTIC_WRITE)` and makes others wait. I use it for short, high-contention updates."

8.15 Open Session in View

What it is

Open Session in View (OSIV) keeps the persistence context open for the **whole web request**, including controller code and JSON serialization, not just during the service transaction. Boot enables it by default for web applications (`spring.jpa.open-in-view=true`) and logs a warning at startup:

```
WARN  JpaBaseConfiguration$JpaWebConfiguration - spring.jpa.open-in-view is enabled by default. Therefore, database queries may be performed during view rendering. Explicitly configure spring.jpa.open-in-view to disable this warning
```

The demo controller touches a lazy collection outside any `@Transactional` method:

```
@GetMapping("/authors/{id}/summary")
String summary(@PathVariable Long id) {
    Author a = authors.findById(id).orElseThrow();
    return a.getName() + " has " + a.getBooks().size() + " books";   // lazy load in the controller
}
```

```
open-in-view=true  -> 200 Terry Pratchett has 3 books
open-in-view=false -> Cannot lazily initialize collection of role 'ch08.jpa.Author.books' with key '2' (no session)
```

Trade-offs

OSIV on	OSIV off
lazy loading "just works" anywhere in the request	lazy loading only inside transactions
a database connection may be held for the whole request, including slow serialization	connections held only during transactions
lazy loads during serialization run in auto-commit mode, one query each (hidden N+1)	forces you to fetch explicitly per use case
convenient for prototypes	predictable and recommended for APIs

Recommendation: set `spring.jpa.open-in-view=false` in new applications, fetch what each use case needs inside the service, and return DTOs. The failing test above is exactly what you want to see during development rather than in production under load.

8.16 Schema migrations with Flyway

What it is

In production you never let Hibernate change the schema. A **migration tool** applies small, numbered SQL scripts in order, **exactly once per database**, and records each one in a history table. The schema's history then lives in version control, like code. **Flyway** (SQL files) and **Liquibase** (XML/YAML/SQL changelogs) are the two Boot supports.

Boot 4 change: Flyway is only auto-configured when you use `spring-boot-starter-flyway` . `flyway-core` on its own is no longer enough, because of Boot 4's modular auto-configuration. For PostgreSQL also add `org.flywaydb:flyway-database-postgresql` (database support has been split into separate modules since Flyway 10).

```
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-flyway</artifactId>
</dependency>
```

```
src/main/resources/db/migration/
  V1__create_customer.sql        CREATE TABLE customer (id ... PRIMARY KEY, name VARCHAR(100) NOT NULL);
  V2__add_customer_email.sql     ALTER TABLE customer ADD COLUMN email VARCHAR(200); CREATE UNIQUE INDEX ...
  V3__seed_customers.sql         INSERT INTO customer (name, email) VALUES ('Asha', ...), ('Ben', ...);
```

```
spring.jpa.hibernate.ddl-auto=validate   # Hibernate only checks the entities against Flyway's schema
```

```
@DataJpaTest(showSql = false)            // the JPA slice includes Flyway
class MigrationTest {
    @Test
    void migrationsRanInOrderAndHibernateValidatedTheSchema() {
        for (MigrationInfo m : flyway.info().all())
            System.out.printf("V%s  %-25s %s%n", m.getVersion(), m.getDescription(), m.getState());
        // + print flyway_schema_history and the customers
    }
}
```

```
V1  create customer           SUCCESS
V2  add customer email        SUCCESS
V3  seed customers            SUCCESS
history: -1 | null | << Flyway Schema History table created >> | TABLE
history: 1 | 1 | create customer | SQL
history: 2 | 2 | add customer email | SQL
history: 3 | 3 | seed customers | SQL
customer: Asha <asha@example.com>
customer: Ben <ben@example.com>
```

(Flyway 12.4 also printed a warning that H2 2.4 is newer than the H2 version it was verified with, which is harmless here.)

How it works

1. At startup, before JPA initializes, Boot runs Flyway against the `DataSource`.
2. Flyway reads `flyway_schema_history`, finds the scripts not yet applied, runs them in version order (each in a transaction where the database allows), and records each with a **checksum**.
3. Hibernate starts with `ddl-auto=validate`. If an entity has a column the schema lacks, for example if V2 were missing, startup fails. That is a useful safety net.

Rules

Rule	Why
Never edit a migration that has been applied anywhere	Flyway's checksum check fails (`validate` error). Fix forward with a new version
Naming: `V<version>__<description>.sql` (two underscores), `R__<name>.sql` for repeatable scripts (views, functions)	how Flyway orders and classifies
Make changes backward-compatible (expand → migrate → contract)	during a rolling deploy, old and new app versions share one schema
Big data changes in separate, batched migrations	one huge transaction locks tables
Test migrations against the real database type	H2 accepts SQL that PostgreSQL rejects, and the other way round. Tests with PostgreSQL via Testcontainers: ch12

8.17 Testing JPA code

`@DataJpaTest` (package `org.springframework.boot.data.jpa.test.autoconfigure` in Boot 4, from `spring-boot-starter-data-jpa-test`) starts only JPA: entities, repositories, `EntityManager`, `TestEntityManager`, an embedded database, and Flyway/Liquibase if present. Like `@JdbcTest`, it wraps each test in a rolled-back transaction and turns on `show-sql` (set `showSql = false` to silence it).

Remember what that transaction hides. Inside it, lazy loading always works and nothing is ever committed. That is why this chapter's `LazyLoadingTest`, `AuditingAndLockingTest` and `OpenInViewTest` use `@SpringBootTest` **without** a test transaction. Call `em.flush()` and `em.clear()` in `@DataJpaTest` tests when you want to see the SQL and read back from the database, not from the cache. More in ch12.

8.18 Performance checklist

Check	How
All `@ManyToOne` / `@OneToOne` LAZY	`fetch = FetchType.LAZY`
No N+1 on hot paths	assert statement counts in tests. Use JOIN FETCH / `@EntityGraph`
Safety net	`spring.jpa.properties.hibernate.default_batch_fetch_size=16` (or similar)
Read-only screens	projections / DTOs, `@Transactional(readOnly = true)`
Batched inserts	SEQUENCE ids + `hibernate.jdbc.batch_size=50` + `hibernate.order_inserts=true` (IDENTITY disables insert batching)
OSIV off	`spring.jpa.open-in-view=false`
No unbounded queries	paging (keyset for deep pages), `Limit`
Big batch jobs	chunk + `flush()` / `clear()`, or JDBC
Indexes for your query patterns	foreign keys are not indexed automatically in PostgreSQL. See SQL ch12

8.19 Interview questions

1. **JPA vs Hibernate vs Spring Data JPA?** → Specification vs implementation vs Spring's repository layer on top.
2. **What is the persistence context?** → The `EntityManager`'s first-level cache and unit of work. One instance per row, snapshots, dirty checking, write-behind at flush.
3. **Why don't I need `save()` after changing a loaded entity?** → Dirty checking at flush, inside a transaction.
4. **Entity states?** → Transient, managed, detached, removed.
5. **What is the owning side?** → The side with the foreign key (`@ManyToOne`). JPA only reads it. The other side uses `mappedBy`.
6. **`orphanRemoval` vs `CascadeType.REMOVE`?** → Removal from the collection deletes the child vs deleting the parent deletes the children.
7. **Default fetch types?** → To-one: EAGER. To-many: LAZY. Make everything LAZY.
8. **What causes `LazyInitializationException` and how do you fix it?** → Touching a lazy association after the persistence context closed. Fetch it in the query, work inside a transaction and return DTOs. Not EAGER, and preferably not OSIV.
9. **Explain N+1 and three fixes.** → 1 query + 1 per parent. JOIN FETCH, `@EntityGraph`, batch fetching (and DTO projections).
10. **JOIN FETCH vs `@EntityGraph`?** → The same idea. JOIN FETCH is written in JPQL (inner join unless `left`). An entity graph is declared on the method and uses a left join.
11. **How does `save()` choose between persist and merge?** → `isNew`: null id, or null wrapper version. Otherwise merge.
12. **Derived query vs `@Query` vs native?** → Short conditions vs readable JPQL vs database-specific SQL.
13. **`Page` vs `Slice`?** → `Page` runs an extra COUNT and knows totals. `Slice` only knows whether there is a next one.
14. **Interface vs DTO projections?** → Both select fewer columns. Records are plain values, interfaces are proxies. Open interface projections load the entity.
15. **How does optimistic locking work?** → A `@Version` column, `WHERE version = ?`, 0 rows → `ObjectOptimisticLockingFailureException`.
16. **What is OSIV and should it be on?** → The persistence context stays open for the whole web request. It is on by default with a warning. Turn it off for APIs.
17. **`ddl-auto` in production?** → `validate` or `none`, with Flyway/Liquibase owning the schema.
18. **What changed for Flyway in Boot 4?** → Use `spring-boot-starter-flyway`. `flyway-core` alone is not auto-configured.

Chapter TL;DR

- **JPA** is the spec, **Hibernate 7** the implementation, **Spring Data JPA** the repository layer. Boot 4.1 ships Hibernate 7.4 and Spring Data 2026.0.
- The **persistence context** gives one object per row, tracks changes, and flushes `UPDATE`s automatically (**dirty checking**). Detached entities are not tracked.
- **Relationships**: the owning side holds the foreign key. Use `mappedBy` on the other side and keep both in sync. Cascade from aggregate roots only. `orphanRemoval` deletes removed children.
- **LAZY everything.** `LazyInitializationException` means you touched it outside a transaction. Fetch per use case instead.
- **N+1** is 1 + N queries, measured here as 5 statements for 4 authors. Fix it with `JOIN FETCH` (1), `@EntityGraph` (1), batch size (2), or projections. EAGER makes it worse, not better.
- **Repositories**: derived queries, `@Query` (JPQL/native), `@Modifying` (clears nothing unless asked), `Page` / `Slice` / `Window`, projections for reads.
- **Auditing** fills created/modified columns. `@Version` turns lost updates into `ObjectOptimisticLockingFailureException`. `@Lock(PESSIMISTIC_WRITE)` gives `FOR UPDATE`.

- Turn **open-in-view off**, and let **Flyway** (`spring-boot-starter-flyway` in Boot 4) own the schema with `ddl-auto=validate`.

Next: Chapter 9 — Spring Data beyond JPA, and caching covers Spring Data JDBC, MongoDB, Redis and the cache abstraction.